

The University of Hong Kong

Department of Geography

Course code and title:

GEOG7305 Programming for Geospatial Data

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Type of assignment (e.g. essay, practical exercise, etc.):

Lab01 Data Flow Diagram

Project Title: HKU Smart Library Seat Management System

1. Project Proposal & Reasons

1.1 Proposal

A real-time seat booking and monitoring system integrated with HKU Student ID and IoT sensors.

1.2 Reasons

This system cloud optimize the utilization of library study spaces and prevent prolonged seat-hogging behavior,and reduce unnecessary energy consumption.

2. System Description

The HKU Smart Library Seat Management System is a smart tool designed to help students find and book study seats easily. Following the SSADM method, the system focuses on how information moves between users and the database,we could see the process starts when a student sends a Booking Request, which includes their ID and the seat they want to choose and the system uses the Identity Verification process to check the student's information in the Student Info database(D1),which ensures the student is allowed to use the library.

After first process, the Seat Monitoring process receives signals from sensors on the library seats and this information could be saved in the Seat Status database(D2) that make the smart system always know which seats are empty. Finally, the system connects the student's request with the seat data to send a Booking Confirmation.This system makes the library more efficient and helps HKU become a greener and smarter campus.

3. System Specification

(1) Inputs

Student ID, Seat Selection, Booking Time, Sensor Signals.

(2) Outputs

Booking Confirmation,Booking Status, Management Reports for Admin.

(3) Data Stores

D1 | Student Info

Holds student credentials and eligibility status.

D2 | Seat Status Table

Stores seat IDs, current states (Available/Occupied),Expiry Time.

4. Data Flow Diagrams

4.1 Context Diagram

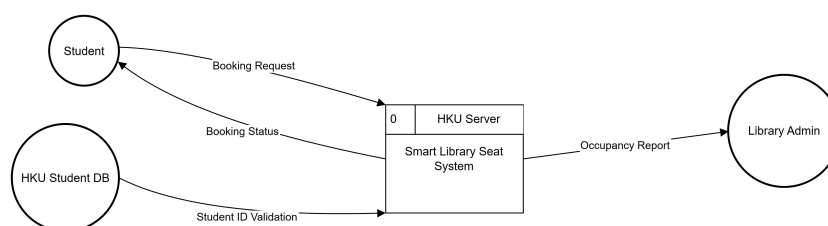


Figure 1. Context-level DFD representing the system boundary.

The context diagram shows the overall interaction between the system (Process 0) and external entities like Students and HKU Student DB.

4.2 Level 1 Diagram

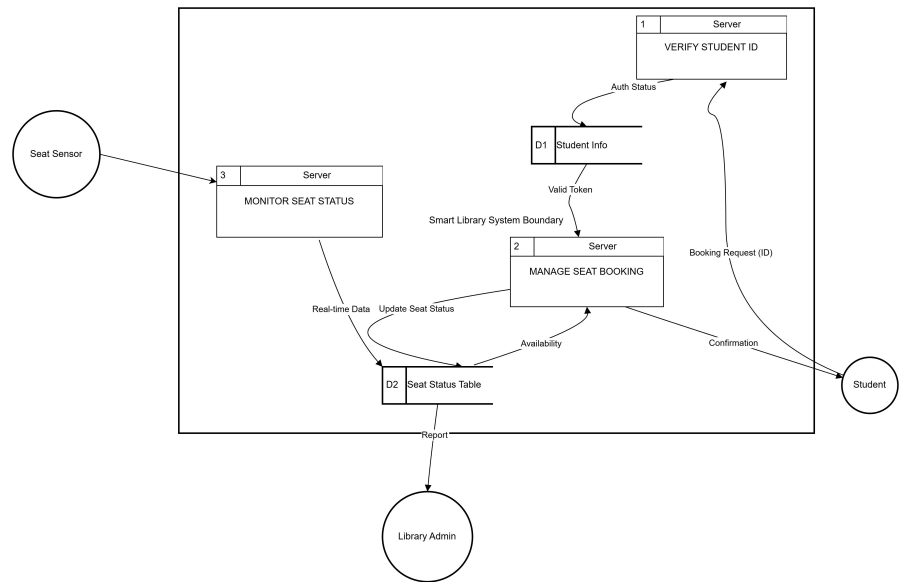


Figure 2. Level 1 DFD decomposing the system into primary functional processes.

The level 1 diagram illustrates how data flows between identity verification, booking management, and sensor monitoring activities.

4.3 Level 2 Diagram

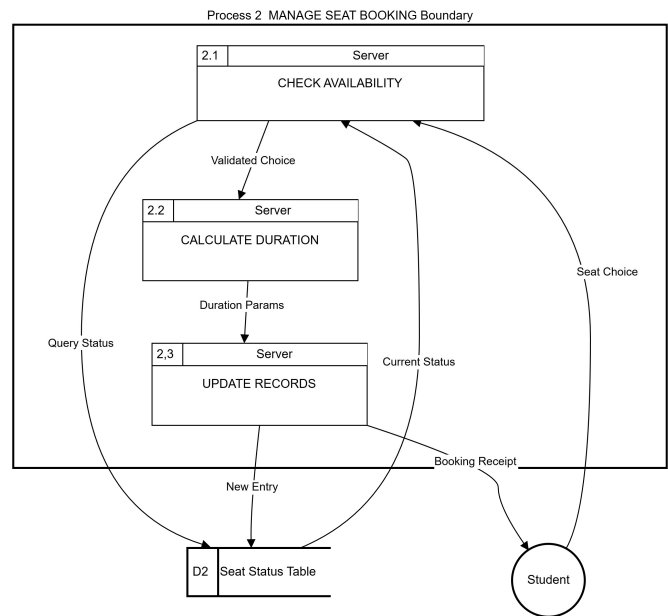


Figure 3. Detailed breakdown of Activity 2: MANAGE SEAT BOOKING.

This level 2 diagram reveals the internal sub-processes (2.1, 2.2, 2.3) used to handle a specific booking transaction from the manage seat booking process.

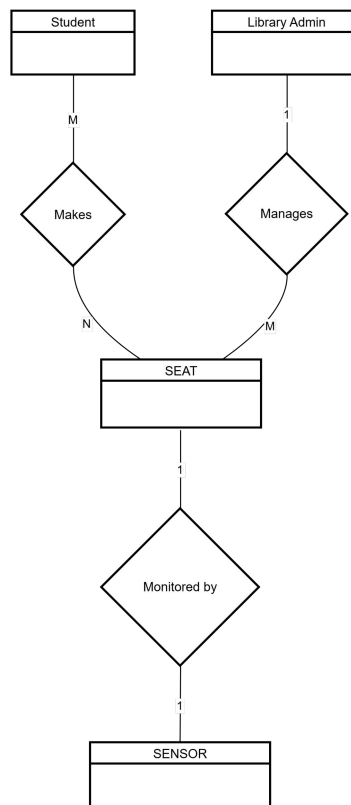
5.1 Exercise 1: Draw ER-Diagram

Ex1.1 List out the Entities:

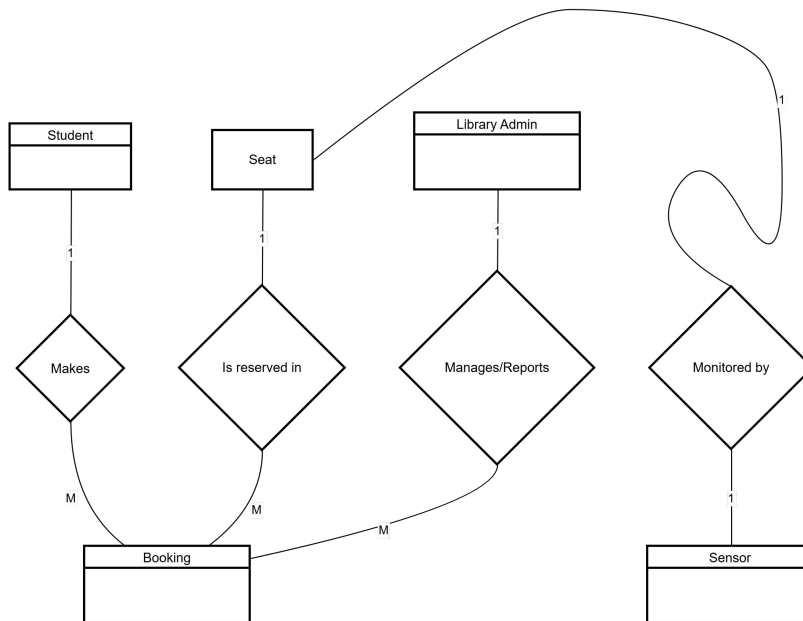
Identify Relationships (fill in the blank with *):

	Student	Seat	Sensor	Booking	Library Admin
Student				*	
Seat			*	*	
Sensor		*			
Booking	*	*			*
Library Admin				*	

Ex1.2 Draw the entities and relationship (rough)



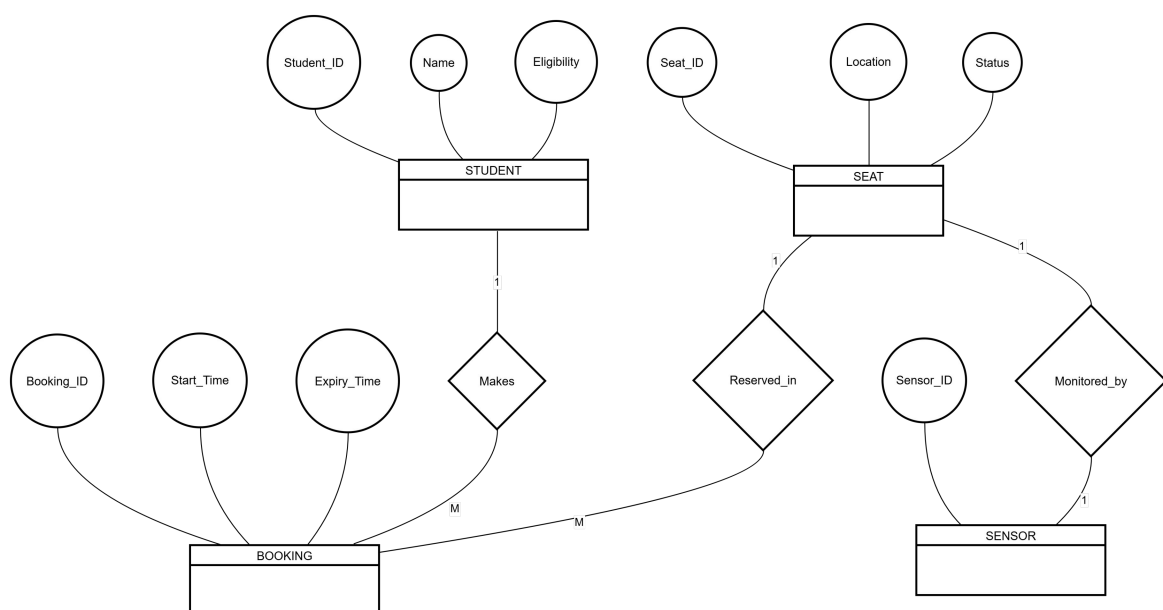
Ex1.3 Eliminate many-many relationships, and collapse one-to-one relationships



Ex1.4 Define Primary Keys (fill in the blank)

Entity	Primary Key
Student	Student_ID
Library Admin	Admin_ID
Seat	Seat_ID
Sensor	Sensor_ID
Booking	Booking_ID

Ex1.5 Full ER diagram(fill in the cardinality and attribute)



The ERD design for the HKU Smart Library Seat Management System aims to structure the logical data flows identified in the Lab 01 DFD. By introducing 'Booking' as a bridge entity, the many-to-many relationship between 'Student' and 'Seat' has been successfully resolved, which ensures data consistency. Furthermore, the one-to-one relationship between 'Sensor' and 'Seat' allows for the continuous, real-time monitoring of seat occupancy status. All entities are assigned unique primary keys, such as Student_ID and Seat_ID, laying a firm foundation for the eventual physical implementation of the database.

Ex1.6 Finally Define Database Tables:

Student

<u>Student ID</u>	Student Name	Eligibility Status
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Seat

<u>Seat ID</u>	Location	Current Status(Available/Occupied)
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Booking

<u>Booking ID</u>	Student ID	Seat ID	Booking Time	Expiry Time
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Sensor

<u>Sensor ID</u>	Seat ID	Sensor Signal
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Library Admin

<u>Admin ID</u>	Admin Name	Department
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